

Class Activity: Software Problem Analysis and System Modeling

Short Description

Think of a small software system that solves a real business or social problem. Describe the problem, propose a software solution, identify functional and non-functional requirements, select a suitable software development model, and draw five system diagrams.

Student Task

Choose one software idea and complete the following:

1. Problem Description

Describe the problem, target users, and why it is important.

2. Proposed Software Solution

Explain the software system and how it solves the problem.

3. Functional Requirements

Write at least 5 functional requirements (what the system should do).

4. Non-Functional Requirements

Write at least 5 non-functional requirements (how the system should perform).

5. Development Model

Choose a model (Waterfall, Agile, Prototype, Spiral) and justify your choice.

Required Diagrams

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram
- Deployment or Component Diagram

Submission Format

Submit a document including:

1. Title
2. Problem Description
3. Solution
4. Functional Requirements
5. Non-Functional Requirements
6. Development Model
7. Diagrams